

Home Appliance Energy Consumption and Efficiency Guide

Introduction

Household appliances account for a significant portion of residential electricity consumption. Understanding appliance power ratings helps consumers estimate energy usage and operating costs.

Understanding Power Ratings

Power ratings indicate the rate at which electrical energy is consumed.

Power is measured in watts (W).

Common household appliance ratings:

Appliance	Typical Power
LED Bulb	9 W
Ceiling Fan	75 W
Refrigerator	150 W
Television	120 W
Washing Machine	500 W
Air Conditioner	1500–2000 W

Energy Consumption Formula

Energy (kWh) = Power (kW) × Time (hours)

LED Lighting Example

LED Lamp Power = 9 W

Operating Time = 8 hours/day

Energy:

0.009×8

= 0.072 kWh/day

Monthly Energy:

0.072×30

= 2.16 kWh/month

Ceiling Fan Example

Fan Power = 75 W

Usage = 10 hours/day

Energy:

$$0.075 \times 10$$

$$= 0.75 \text{ kWh/day}$$

Monthly Consumption:

$$0.75 \times 30$$

$$= 22.5 \text{ kWh/month}$$

Refrigerator Consumption

A refrigerator cycles on and off automatically.

Typical Annual Consumption:

150–400 kWh/year

Example:

Annual Consumption = 300 kWh

Monthly Consumption:

$$300 \div 12$$

$$= 25 \text{ kWh/month}$$

Air Conditioner Consumption

Air conditioners are typically the largest residential energy consumers.

Example:

AC Rating = 1.8 kW

Daily Usage = 8 hours

Energy:

$$1.8 \times 8$$

$$= 14.4 \text{ kWh/day}$$

Monthly Usage:

$$14.4 \times 30$$

$$= 432 \text{ kWh/month}$$

Monthly Operating Cost Example

Electricity Rate:

₹8/kWh

AC Consumption:

432 kWh/month

Cost:

$432 \times ₹8$

= ₹3456/month

Energy Efficiency Ratings

The Bureau of Energy Efficiency (BEE) provides star ratings.

Higher star ratings generally indicate lower electricity consumption.

Star Ratings:

- 1 Star
- 2 Star
- 3 Star
- 4 Star
- 5 Star

A 5-star appliance typically consumes less energy than a comparable 1-star model under standardized testing conditions.

Standby Power Consumption

Many appliances continue consuming electricity while not actively operating.

Examples:

- Television
- Set-top box
- Wi-Fi router
- Microwave display

This is commonly called phantom load.

Methods to Improve Appliance Efficiency

- Clean AC filters regularly
 - Maintain refrigerator door seals
 - Use full loads in washing machines
 - Turn off appliances completely when not needed
 - Upgrade inefficient appliances
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Comparing Fan and Air Conditioner

Example:

Fan:

75 W

10 hours/day

= 0.75 kWh/day

Air Conditioner:

1800 W

10 hours/day

= 18 kWh/day

The AC consumes substantially more energy than the fan under similar operating durations.

Conclusion

Appliance energy consumption depends on power rating and usage duration. Understanding these factors enables accurate cost estimation and informed purchasing decisions.